Evaluated Under What? A Channel-Realism Audit of Semantic Communication for Non-Terrestrial Networks

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Abstract

Semantic communication is recommended for non-terrestrial networks on the argument that it tolerates the conditions those links impose: large propagation delay, Doppler shifts of hundreds of kilohertz, short visibility windows and an elevation-dependent link budget. Surveys of the area map each of these limitations onto the works said to address it. We ask the narrower question those mappings leave open, namely whether the works cited as addressing a limitation are evaluated under it. Two measurements answer it. First, propagating 1,200 public Starlink two-line element sets with SGP4 over 30,922 passes fixes the magnitude of each effect from orbital geometry rather than from assumption: peak Doppler reaches 359.2 kHz at Ka-band with a drift of up to 16.9 kHz/s, and the geometric term of the link budget sweeps 3.72 dB across a median visibility window of 206.0 s. Second, a coding of the 46 works a recent survey lists as representative, of which 27 could be obtained in full text, records for each whether an effect appears in the framing of the paper or in its evaluation. Of 26 codeable works, 16 evaluate over an additive white Gaussian noise channel, 4 use a standardised non-terrestrial channel model, 0 drive their channel from orbital data, and 0 instantiate the time-varying component of Doppler. One widely reasonable suspicion is not supported and we report it as a result: the link budget moves only 0.049 dB/s, so treating the signal-to-noise ratio as static within a transmission block is sound. The checklist, the coding sheet with the sentence that justifies every decision, the orbital generator and the evaluation harness are released.

Keywords: Semantic communication, Non-terrestrial networks, LEO satellites, Evaluation methodology, Reproducibility, Channel modelling

1. Introduction

Semantic communication proposes to transmit what a message means for a downstream task rather than the symbols that encode it, and it is increasingly recommended for non-terrestrial networks (NTN). The argument is one of fit. Satellite links impose constraints that a conventional pipeline handles poorly, and a recent survey of the area states them explicitly [1]: severe and time-varying path loss, long propagation delay, large and time-varying Doppler shift, intermittent connectivity with short visibility windows, atmospheric attenuation, on-board size, weight and power limits, and distortion that accumulates over multiple hops.

That survey does something more specific than list the constraints. It tabulates a mapping from each NTN limitation to the semantic-communication property said to address it, and cites representative works against each pairing. The mapping is a claim about the literature, and it is the claim this paper tests. A work can be cited as addressing Doppler because it discusses Doppler, and can nevertheless be evaluated over a channel in which no Doppler is present. Whether that gap exists, and how wide it is, has not been measured.

What this paper does.. Figure 1 sets out the audit. Two measurements are taken separately and neither depends on the other. The first establishes what the orbit does, by propagating public two-line element sets and computing the quantities an NTN-specific evaluation would have to instantiate. The second establishes what the literature evaluates, by coding the works a survey lists as representative against a checklist anchored to the 3GPP non-terrestrial channel studies [2] rather than to criteria of our own devising. A third measurement then asks what the gap costs, by training a semantic codec under the modal assumption of the population and evaluating it under the conditions the first measurement found.

What we do not claim.. Three boundaries are set at the outset because an audit is easy to overstate. We do not claim the field is careless: the survey's own tables carry a channel-consideration column, so the community records this information, and our contribution is that nobody has compared the record against the evaluations. We propose no new semantic-quality metric, an area addressed by recent work that approaches evaluation from the metric side rather than the conditions side [3]. And we do not revisit whether learned codecs beat classical separation-based baselines, a question the deep joint source-channel coding literature already treats carefully [4, 5], including published cases where the classical baseline wins.

A finding that runs the other way.. One suspicion motivates a natural version of this audit and turns out to be wrong. If the link budget swept quickly, a fixed signal-to-noise ratio would be indefensible even within a single image transmission. Section 4 measures the sweep at 0.049 dB/s, which over ten seconds is 0.49 dB. Treating the channel as static within a transmission block is therefore sound, and we report this because it is the first objection a reader should raise and because it narrows the audit to the effects that survive measurement.

2. Background

2.1. Why non-terrestrial links are said to be different

A satellite in low Earth orbit moves at roughly 7.5 km/s relative to the ground, which is the origin of every constraint in the list above. Motion at that speed produces a Doppler shift proportional to the carrier frequency, so a link that is untroubled at S-band becomes demanding at Ka-band. It also bounds the time a given satellite is usable, since a pass ends when the elevation angle falls below the minimum the link budget supports. And because free-space path loss depends on slant range, which varies over the pass, the received power is not constant even in the absence of fading.

These quantities are geometric. They follow from orbital elements, a ground station position and a carrier frequency, and they can be computed rather than assumed. Section 4 does so, which is what allows the rest of the paper to speak about magnitudes instead of adjectives.

2.2. Semantic communication and its evaluation

Learned joint source-channel coding maps a source directly to channel symbols with a neural encoder and reconstructs it with a neural decoder, trained end to end through a differentiable channel [4]. The shift of objective from reproducing symbols to preserving meaning is a departure

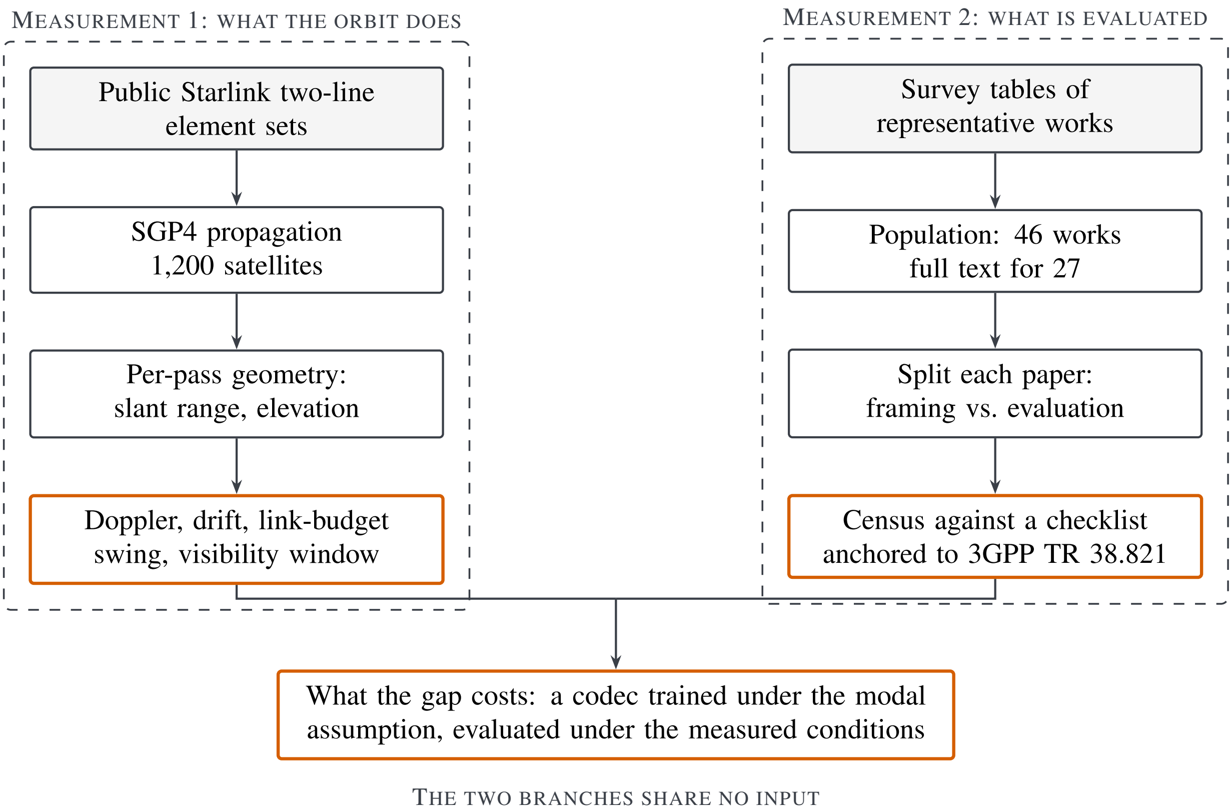

Figure 1: The audit. The two measurements are independent: the orbital side never consults the literature and the census never consults the orbit. They meet only in Section 6, where the conditions measured on the left are applied to a codec trained under the assumption counted on the right.

from the classical formulation [6] and is set out in [7]. The channel is therefore not an external assumption but a component of the training objective, which is why the choice of channel model has consequences that reach beyond the reported numbers: a codec learns to invert the impairments it is trained against and has no reason to handle those it never saw.

Evaluation practice in the area has begun to receive attention from the metric side, asking how semantic success should be defined and measured across modalities and tasks [3]. The present paper is complementary and asks about conditions rather than metrics: given any metric, under what channel was it computed.

2.3. Audits as a genre

Reassessments that re-run a body of work under stricter conditions have an established place in networking. They have found reported gains in traffic classification that arose from dataset-specific shortcuts rather than task semantics [8], and reality checks in adjacent areas have been published even when the reassessed method loses [9], because the contribution is the evaluation protocol rather than a new mechanism. This paper follows that genre. Its contribution is a protocol, a measurement of what the protocol is measuring against, and a released harness.

3. The audit protocol

3.1. Population

The population is the set of works a recent survey of semantic communication for non-terrestrial networks [1] lists in its representative-works tables, covering satellite, aerial and integrated scenarios. Using a survey as the sampling frame is a convenience sample and we treat it as one, with the consequences discussed in Section 8. It has one property that outweighs the cost: the pairing between an NTN limitation and the works said to address it is made by the field rather than by us, so the audit tests a claim the community has already put in print.

The tables list 46 distinct works [10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36]. Full text could be obtained for 27, which is 58.7% of the population; the remaining 19 are counted as not auditable rather than guessed at, since a work whose evaluation section cannot be read is a fact about the accessibility of the literature and not a gap to be filled by inference.

3.2. Checklist

Table 4 states the checklist: 12 items covering the physical effects an NTN evaluation could instantiate and the channel models it could use, each with the source it is anchored to. 7 of the items restate structural constraints the field's own survey names, the standardised-model item points at the 3GPP studies, and the remainder are protocol conventions rather than physical effects. Two design decisions matter. The effects are those the field's own survey names as structural NTN constraints, and the channel-model entries follow the 3GPP non-terrestrial studies [37, 2], so neither list is ours. And the checklist is deliberately about *conditions of evaluation*, not about quality: a work that evaluates over additive white Gaussian noise is not thereby wrong, it is evaluated under a stated condition, and the audit records which condition.

3.3. Coding, and where a claim counts

The instrument is a distinction between two regions of a paper. A sentence about Doppler in an introduction or a related-work section states a motivation. The same sentence in a simulation setup or results section is evidence that the evaluation instantiates the effect. Every paper is therefore split at the first heading that begins its evaluation, and every checklist item is coded separately for the region before and the region after.

The extraction step returns sentences rather than verdicts. For each work and each checklist item, the harness collects the sentences that bear on the item, tagged by region, and a decision is recorded against those sentences. The coding sheet released with this paper carries the sentence behind every decision, so any individual judgement can be disputed without repeating the collection.

3.4. Testing the instrument before trusting it

Two checklist items return zero across the whole population, and a zero is exactly where a broken detector is indistinguishable from a finding. Three controls therefore run before any zero is quoted. Sentences written the way papers actually write them must be caught, including phrasings that avoid the obvious keyword. Near misses must be rejected, among them a mention of 3GPP that is not a channel model. And the detectors must demonstrably fire on the real corpus, which rules out an empty-input failure in which every count is zero because nothing was read. All three pass, and the controls are released as an executable script rather than described.

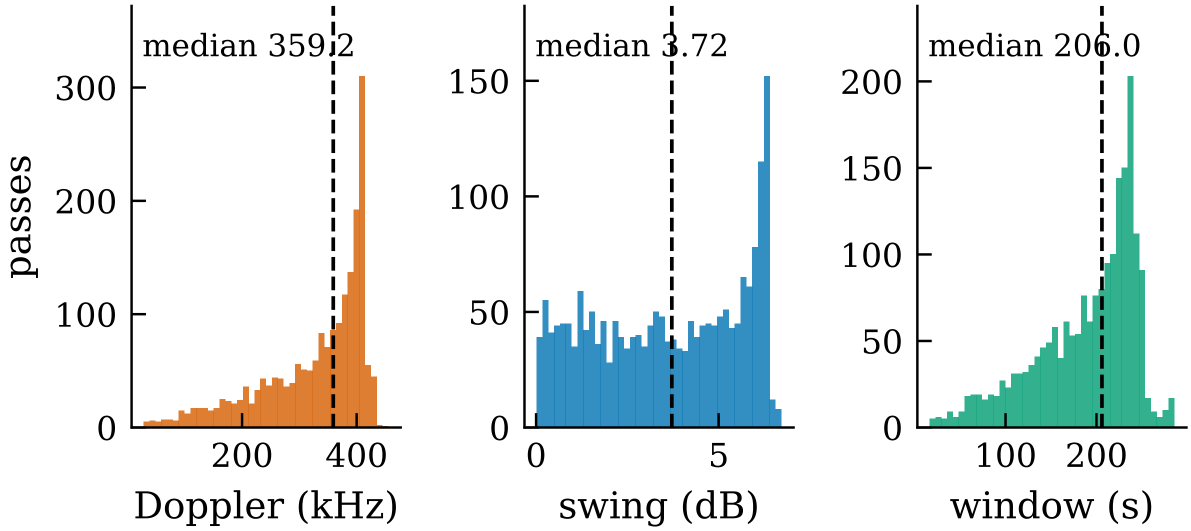

Figure 2: Distributions over 1,978 passes at one station with a minimum elevation of 25°: peak Doppler shift at Ka-band, the swing of the geometric link-budget term across a pass, and the visibility window. Dashed lines mark medians. The bimodality of the centre panel reflects the difference between overhead passes, which sweep the full range of slant range, and grazing passes, which do not.

4. What the orbit actually does

4.1. Method

A seeded sample of 1,200 Starlink two-line element sets is propagated with SGP4 [38] over 30,922 passes at four ground stations spanning 0 to 65 degrees of latitude. For each pass we record slant range, elevation, one-way delay, the Doppler shift implied by the range rate at S-band and Ka-band, its time derivative, and the free-space path loss.

Only geometry, deliberately.. We compute free-space path loss and nothing else. Antenna pattern roll-off across a pass, atmospheric and rain attenuation and receiver noise figure all *add* variation rather than removing it, so the swing reported here is a lower bound on how much the link budget moves. A lower bound requires no contested modelling, and the argument of this paper needs only a lower bound.

4.2. Results

Figure 2 shows the three distributions that matter and Table 1 reports them per station. Peak Doppler at Ka-band has a median of 359.2 kHz and reaches 455.5 kHz, which is the literal content of the phrase "hundreds of kilohertz" that motivates the area. At S-band the same passes give 35.9 kHz. The drift has a median of 5.4 kHz/s and a maximum of 16.9 kHz/s, so the shift is not a constant offset that a single estimate removes. The geometric term of the link budget sweeps a median of 3.72 dB and up to 6.72 dB over a median visibility window of 206.0 s, and one-way delay reaches 3.25 ms.

Table 1: Medians per ground station at a minimum elevation of 25°. The quantities differ by at most 15.6 kHz in Doppler across 65 degrees of latitude, so they are properties of the orbital shell rather than of a chosen site.

Ground station	Lat. (°)	Passes	Window (s)	Swing (dB)	Doppler (kHz)
Da Nang	16.0	1,978	206.0	3.72	359.2
Equator	0.0	1,878	205.5	3.67	356.3
Mid-lat	45.0	3,663	200.5	3.61	354.5
High-lat	65.0	927	215.0	3.87	370.1

Table 2: Full statistics at one station for the three elevation thresholds. Link-budget figures are the geometric term only and are therefore lower bounds.

	10°		25°		30°	
Quantity	median	max	median	max	median	max
Visibility window (s)	370.2	502.0	206.0	285.5	172.0	241.0
One-way delay (ms)	5.41	6.26	3.25	3.90	2.86	3.45
Peak Doppler, S-band (kHz)	38.8	49.7	35.9	45.6	34.3	43.4
Peak Doppler, Ka-band (kHz)	387.8	497.0	359.2	455.5	342.9	433.5
Doppler drift, Ka (kHz/s)	3.66	16.88	5.36	16.88	5.73	16.88
Link-budget swing (dB)	4.97	12.15	3.72	6.72	3.26	5.45
Link-budget slew (dB/s)	0.034	0.143	0.049	0.143	0.052	0.143

4.3. One pass, in detail

Figure 3 follows a single pass, and the pass is chosen by a stated rule rather than for appearance: it is the one whose maximum elevation lies closest to the median of the distribution. Over its 216.0 s the Ka-band shift runs from -352.4 kHz to 353.1 kHz, an excursion of 705.5 kHz that passes through zero at closest approach, so the sign of the offset reverses within a single contact. The shaded band marks a one-second transmission block at the same scale. Its width against the pass is the visual form of the two results of this section: nothing of consequence happens to the link budget inside the block, and a great deal happens across the pass that contains it.

4.4. Sensitivity to the elevation threshold

The minimum elevation a link supports is a design choice, and it moves every quantity above. Figure 4 and Table 2 report the three thresholds the 3GPP studies use. Raising the threshold from 10° to 30° cuts the number of usable passes at one station from 3,662 to 1,672 per day while shortening the window, so a stricter threshold buys a better link at the cost of contact time. The Doppler extremes fall because the geometry excludes the low-elevation part of the pass where the range rate is largest, but they remain in the hundreds of kilohertz at every threshold.

4.5. The result that runs against the audit

The rate at which the link budget moves has a median of 0.049 dB/s. Over a one-second transmission the signal-to-noise ratio therefore shifts by well under a tenth of a decibel, and over ten seconds by 0.49 dB. The common practice of holding the signal-to-noise ratio fixed within a transmission block is sound, and an audit that attacked it would be attacking the wrong thing.

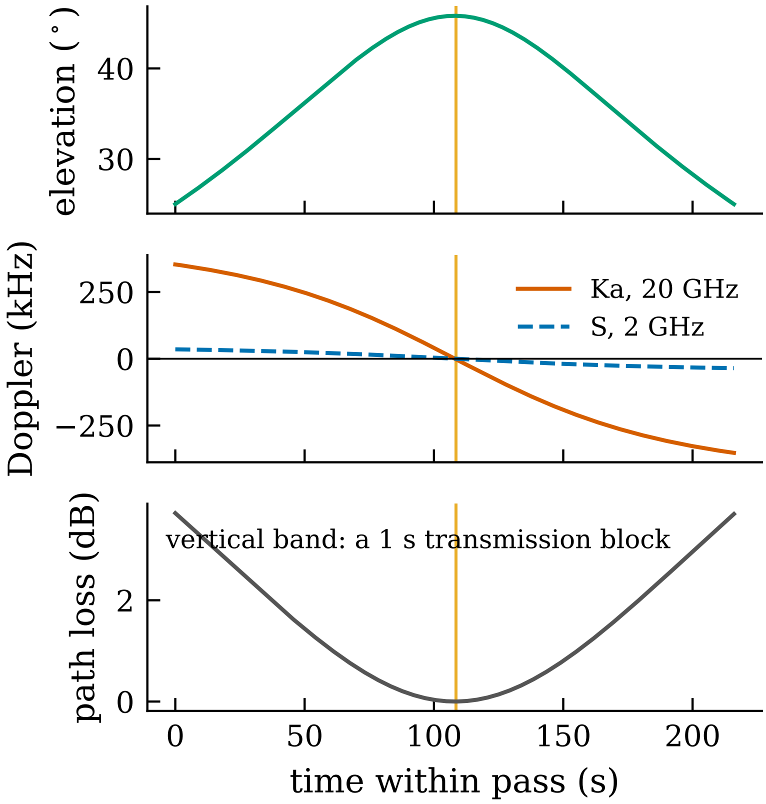

Figure 3: One pass of Starlink 5768 over the station, reaching 45.8°. The selection rule is stated in the text and in the released code. Path loss is plotted relative to its minimum.

What this leaves is a different question, and it is the one Section 6 asks. The sweep is small per second but not small per pass. A codec trained at one operating point is deployed across a window of 206.0 s over which the geometric term alone moves 3.72 dB.

5. What the literature actually evaluates

Figure 6 and Table 5 report the census over the 26 works for which full text was available and an evaluation section could be located.

Channel models.. Additive white Gaussian noise is the modal choice, present in the evaluation of 16 works. Statistical fading appears in 8. A standardised non-terrestrial channel model appears in 4. No work in the population drives its channel from orbital data: the count for two-line elements, SGP4, ephemeris or an equivalent propagation tool is 0.

Effects.. Doppler appears in the evaluation of 11 works and in the framing of 2. Its time-varying component appears in 0. Elevation-dependent loss appears in 9, propagation delay in 8, a finite visibility window in 3, and an on-board compute limit in 3.

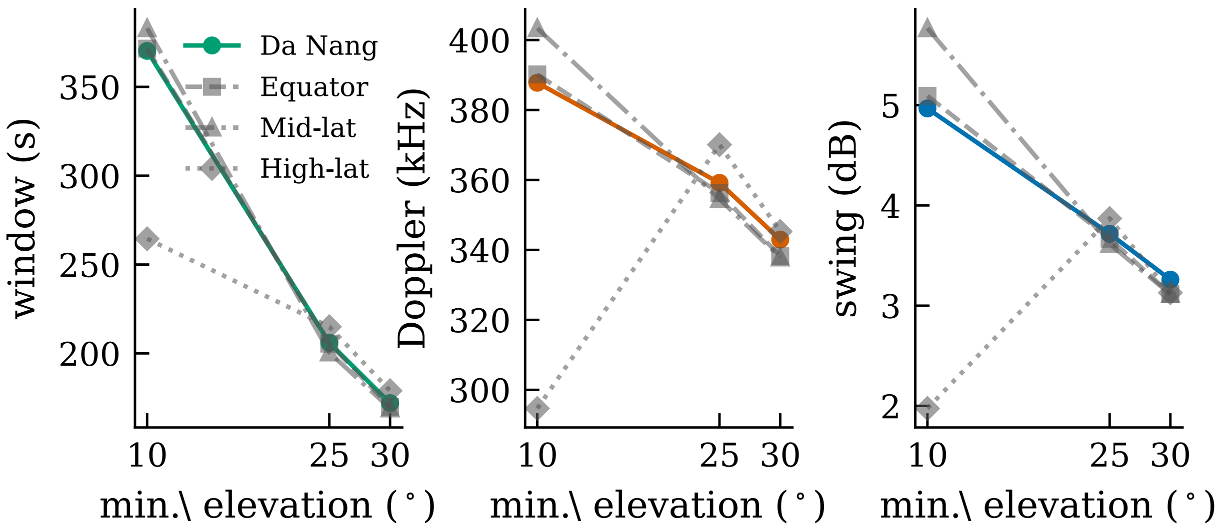

Figure 4: Medians of the same three quantities, visibility window, peak Ka-band Doppler and link-budget swing, against the minimum elevation threshold, for four stations. The station used elsewhere in this paper is drawn in colour and the others in grey; the curves lie close together, which is the same shell property Table 1 reports.

Reading the two zeros.. The zeros for time-varying Doppler and for orbital traces are measurements, not silence: the controls of Section 3 establish that the detectors catch the relevant phrasings and do fire elsewhere in the same corpus. Read against Section 4, they say that the drift the orbit produces, up to 16.9 kHz/s, is instantiated by no evaluation in the population, and that no evaluation is driven by the geometry that produces it.

The structure behind the counts.. Figure 5 resolves the census to individual works. No evaluation in the population instantiates more than 7 of the 12 items, and 2 works instantiate none of them. The two empty rows are the zeros discussed above, and seeing them as rows rather than as bars makes clear that they are not the result of one or two works dominating a count.

A second source describes the same works.. The survey's own channel-consideration column is an independent record of what each work considers, so it can be compared against what the evaluation sections contain. Table 3 does that for the 15 works where both are available. Two things follow, and the first is not a criticism. The column is a phrase of a few words and cannot enumerate an evaluation, so the entries naming fewer items than the evaluations contain is expected. What is informative is what the entries name at all: across 15 works they name 2 checklist items in total, because most entries describe an architecture or a topology rather than a propagation environment. The column is the closest thing the literature has to a record of channel realism, and it is not one. In 1 case the comparison runs the other way, with the column naming an effect the evaluation section does not contain.

6. What the gap costs

The census establishes what the population evaluates and Section 4 establishes what the orbit does. This section joins them by taking the modal assumption of the population, an additive white

Table 3: The survey's channel-consideration entry against the evaluation sections, for the 15 works where both could be read. "Named" counts checklist items the entry mentions, "In eval." counts items the evaluation instantiates.

Work	Survey's channel entry	Named	In eval.	Named but absent
[95]	EO downlink, visibility window	1	0	Finite visibility window
[81]	Multi-hop ISL routing	0	2	—
[93]	Adaptive encoder-decoder	0	6	—
[94]	SNR-aware codec	0	4	—
[100]	Doppler, weather fading	1	6	—
[104]	Hierarchical FL	0	5	—
[113]	On-board sparse encoder	0	1	—
[115]	Multi-gateway hops, SNR-aware	0	4	—
[116]	Cooperative relaying	0	1	—
[117]	Semantic forwarding	0	0	—
[118]	Task-clustered FL	0	1	—
[154]	Inter-satellite links	0	7	—
[161]	Satellite interference	0	2	—
[165]	Direct Satellite Communication	0	4	—
[170]	SNR-aware link	0	4	—

Gaussian noise channel at a single signal-to-noise ratio, and asking what happens to a codec trained under it when the conditions of Section 4 are applied.

6.1. Setup

A convolutional joint source-channel codec [4] is trained end to end through a differentiable complex additive white Gaussian noise channel on a standard image benchmark [39]. Two training regimes are compared. The first fixes the signal-to-noise ratio during training, which is the modal practice in the population. The second randomises it, together with a residual frequency offset, over the ranges Section 4 measured. The second regime is included so that the paper reports a remedy rather than only a deficiency, and so that the cost of the gap can be separated from the cost of fixing it.

The evaluation range is not chosen by us.. The sweep spans the link-budget swing measured in Section 4 rather than a range picked for effect, and because that swing excludes antenna pattern and atmospherics it is the conservative end of what a deployed codec would see.

Residual offset, not raw Doppler.. No deployed system leaves 359.2 kHz uncompensated, so evaluating at raw Doppler would be a straw man. We parameterise instead by the normalised residual offset $\varepsilon = f_{\mathrm{res}}/R_s$, the phase advance added per channel use, which avoids committing to a symbol rate that no work in the population states. A reader substitutes their own R_s to locate their operating point: at the median Ka-band shift of 359.2 kHz and a symbol rate of 10 Msym/s, leaving one per cent of the shift uncorrected puts ε near 3.6×10^{-4}.

7. Recommendations

Three things follow, in decreasing order of confidence.

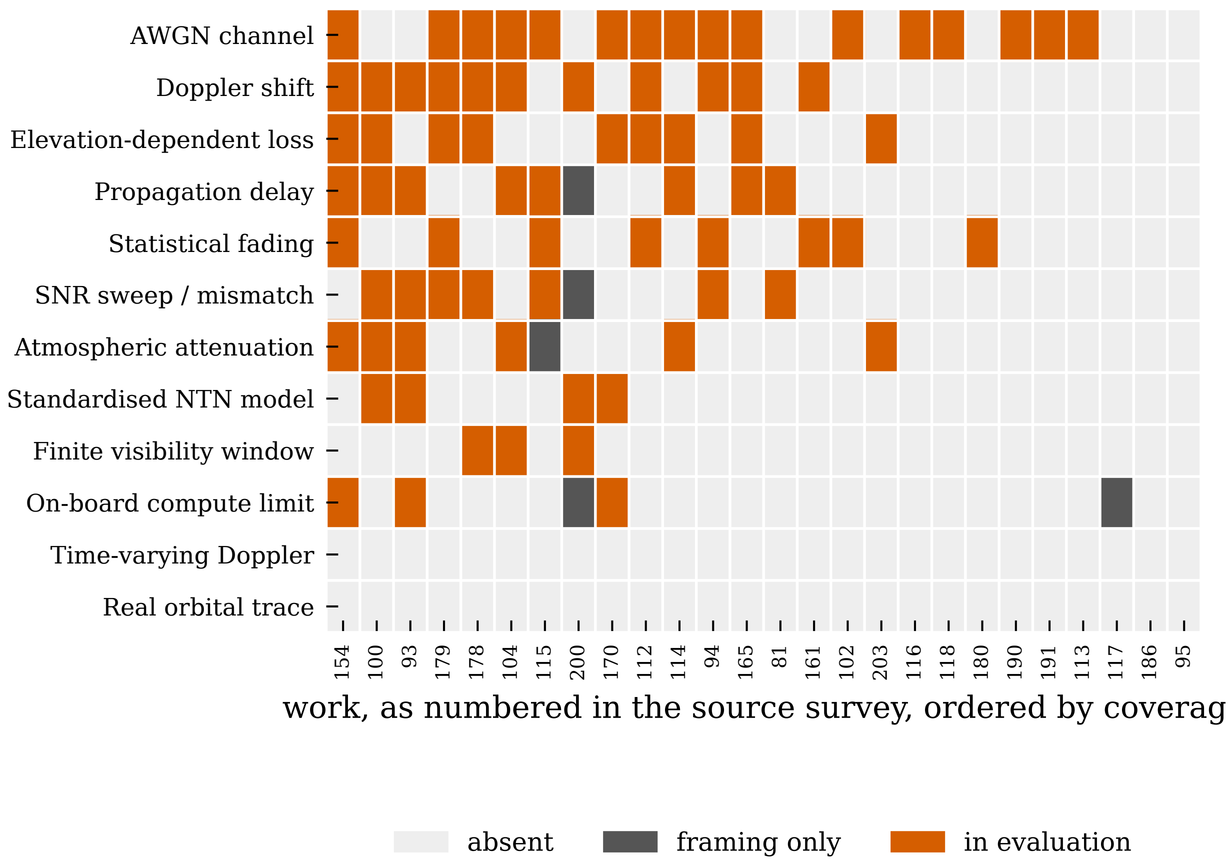

Figure 5: Every codeable work against every checklist item, ordered by coverage. Grey marks an effect that appears only while the problem is being motivated; orange marks one the evaluation instantiates.

State the channel, and state it in the units of the orbit. An evaluation over additive white Gaussian noise at a fixed signal-to-noise ratio is a legitimate experiment and this paper does not argue otherwise. What is missing is the translation: at what elevation, at what point in a pass, and under what residual frequency offset does the stated operating point sit. The generator released here performs that translation from public orbital elements in a few lines.

Report the operating range, not the operating point. A visibility window of 206.0 s over which the geometric term alone moves 3.72 dB is the interval a deployed codec must serve. Reporting a single number at the centre of that interval describes the best case.

Treat residual frequency offset as a swept parameter. Doppler is compensated in practice and no serious system leaves 359.2 kHz uncorrected. What matters is the residual, and because the tolerable residual is a property of the codec rather than of the link, it belongs in the evaluation of the codec.

8. Threats to validity

One survey is one sampling frame. The population is a convenience sample drawn from a single recent survey. It is not a systematic review, and a work absent from that survey is absent

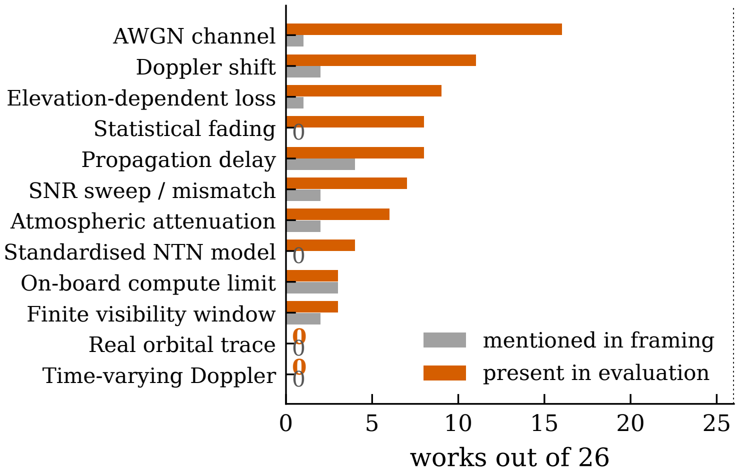

Figure 6: Where each effect appears, for the 26 codeable works. Grey counts works that mention the effect only while motivating the problem; orange counts works whose evaluation instantiates it. Zeros are labelled explicitly so that a measured zero is not read as a missing bar.

here. What the sample buys is that the pairing between limitation and work is the field's rather than ours, which is the specific claim under test.

Two fifths of the population could not be read. Full text was obtained for 27 of 46 works. The 19 that could not be obtained may well instantiate effects the audit does not credit them with. They are excluded from every count rather than assumed absent, and the denominator 26 is stated wherever a proportion appears.

Absence of a phrase is not absence of a model. The census detects the vocabulary in which an effect is normally described. A work could model time-varying Doppler without using any of the phrasings the controls test. This is why the coding sheet is released with the sentences attached rather than only the counts.

The orbital measurement is one constellation. All figures come from Starlink elements at one epoch. Other shells at other altitudes will shift the numbers, though the geometry that produces them does not change in kind.

Free-space path loss is a lower bound, not the link budget. The swing reported here excludes antenna pattern, atmospherics and noise figure. Every one of them widens the range, so the direction of the bound is known even though its tightness is not.

9. Conclusion

Surveys of semantic communication for non-terrestrial networks map each constraint of the satellite link onto the works said to address it. Measuring both sides of that mapping shows where it is load-bearing and where it is not. The orbit produces a Doppler shift with a median peak of 359.2 kHz drifting at up to 16.9 kHz/s and a link budget that sweeps 3.72 dB across a 206.0 s window. Of the 26 works of Section 5 that could be read and coded, 16 evaluate over additive white Gaussian noise, 4 use a standardised non-terrestrial model, and none instantiates either the drift or the geometry that produces it.

Table 4: The checklist, with the source each item is anchored to. "Survey" items restate structural constraints named by the field's own survey, "3GPP" points at the non-terrestrial channel studies, "Orbit" follows from orbital mechanics, and "Protocol" items are evaluation conventions rather than physical effects.

Item	Anchor	What the item asks the evaluation to instantiate
Doppler shift	Survey	Large and time-varying Doppler shift
Time-varying Doppler	Survey	the time-varying component of the same constraint
Propagation delay	Survey	Long propagation delay
Elevation-dependent loss	Survey	Severe and time-varying path loss
Finite visibility window	Survey	Intermittent connectivity, short visibility windows
Atmospheric attenuation	Survey	Atmospheric and weather attenuation
On-board compute limit	Survey	On-board size, weight and power limits
AWGN channel	Protocol	conventional terrestrial reference channel
Statistical fading	Protocol	conventional statistical fading channel
Standardised NTN model	3GPP	TR 38.811 / TR 38.821 non-terrestrial channel
Real orbital trace	Orbit	channel driven by propagated orbital elements
SNR sweep / mismatch	Protocol	operating range rather than operating point

The reading is not that this work is unsound. It is that the case for semantic communication in non-terrestrial networks currently rests on evaluations conducted under terrestrial conditions, and that closing the distance is cheap: the orbital elements are public, the propagation is a library call, and the harness that connects them to an existing codec is released with this paper.

Data availability

The checklist, the coding sheet with the sentence justifying every decision, the orbital generator, the detector controls and the evaluation harness are released.

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Table 5: Census over 26 works. "Framing" counts appearances before the evaluation section, "Evaluation" counts appearances within it. A work may appear in both columns.

Code	NTN effect or channel model	Framing	Evaluation
`CH_AWGN`	AWGN channel	1	16
`DOPPLER`	Doppler shift	2	11
`PATHLOSS_ELEV`	Elevation-dependent loss	1	9
`CH_FADING`	Statistical fading	0	8
`DELAY`	Propagation delay	4	8
`SNR_SWEEP`	SNR sweep / mismatch	2	7
`ATMOS`	Atmospheric attenuation	2	6
`CH_3GPP`	Standardised NTN model	0	4
`ONBOARD`	On-board compute limit	3	3
`VISIBILITY`	Finite visibility window	2	3
`CH_TRACE`	Real orbital trace	0	0
`DOPPLER_RATE`	Time-varying Doppler	0	0

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